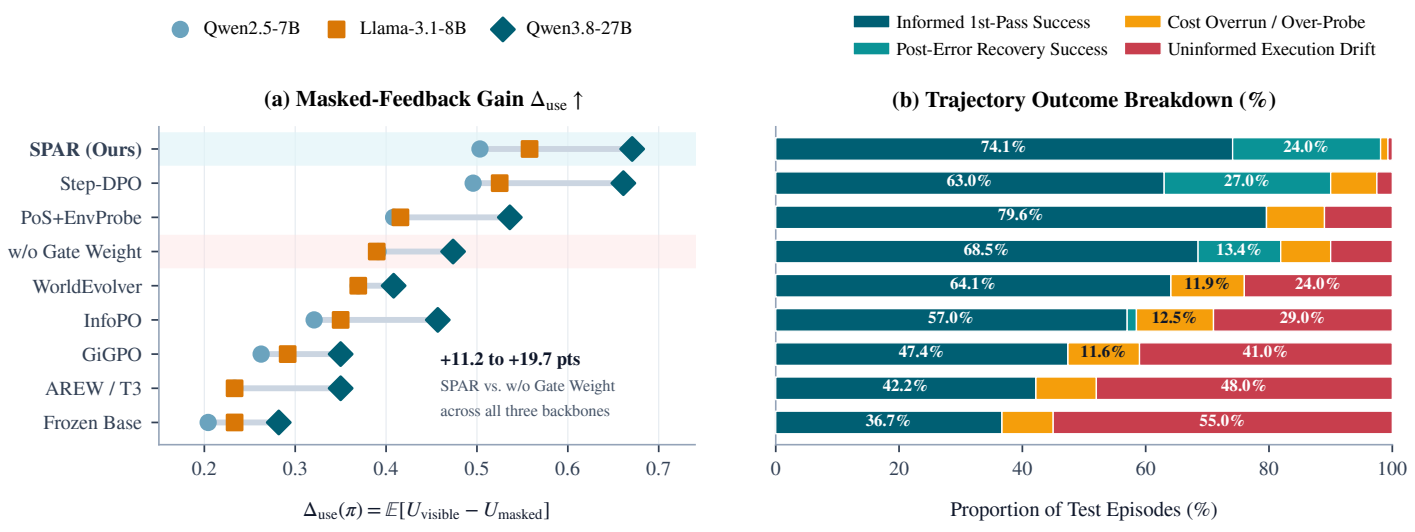
Qwen2.5-7B
Llama-3.1-8B
Qwen3.8-27B
(a) Masked-Feedback Gain Δ_{use} ↑
SPAR (Ours)
Step-DPO
PoS+EnvProbe
w/o Gate Weight
WorldEvolver
InfoPO
GiGPO
AREW / T3
Frozen Base
0.2
0.3
0.4
0.5
0.6
0.7
+11.2 to +19.7 pts
SPAR vs. w/o Gate Weight
across all three backbones
$\Delta_{use}(\pi) = \mathbb{E}[U_{visible} - U_{masked}]$
Informed 1st-Pass Success
Post-Error Recovery Success
Cost Overrun / Over-Probe
Uninformed Execution Drift
(b) Trajectory Outcome Breakdown (%)
74.1%
24.0%
63.0%
27.0%
79.6%
68.5%
13.4%
64.1%
11.9%
24.0%
57.0%
12.5%
29.0%
47.4%
11.6%
41.0%
42.2%
48.0%
36.7%
55.0%
0
20
40
60
80
100
Proportion of Test Episodes (%)